R FOR DATA ANALYSIS · WORKING WITH DATA · LESSON 5

05 Filter rows with filter()

Keep only the rows you need, such as male-headed or female-headed households.

BY THE END OF THIS SECTION YOU WILL BE ABLE TO

- Keep matching rows with filter()
- Use == inside filter, not =
- Save each filtered table with its own name

The R code in this lesson

```
# Filtering a data frame
Male_Only_Demographic <- demographic %>% filter(Gender == "Male")
Female_Only_Demographic <- demographic %>% filter(Gender == "Female")

View(Male_Only_Demographic)
```

What you will hear (times count from the end of the intro)

Intro Filtering is one of the most important skills in data analysis. In this lesson, you will learn to keep only the rows you need with filter, and you will see a very common mistake and how R kindly tells you the fix. Let's filter together.

0:00 I write a note: filtering a data frame. Our goal is a table with only male household heads.

0:10 Male Only Demographic, arrow, demographic, pipe, filter. Inside filter, the condition: Gender equals Male, in quotes.

0:30 Run it. And here is an error. But read it slowly, because R is helping us. It says: we detected a named input. This usually means you used equals instead of double equals. Did you mean Gender, double equals, Male?

0:55 Remember our comparison lesson? One equals sign assigns. Two equals signs compare. Inside filter, we are asking a question, so we need two.

1:25 I add the second equals sign, and run it again. It works! A new table appears in the Environment: one hundred thirty three rows. Only male household heads.

1:40 Let's open it with View, and check the Gender column. Every row says Male. Perfect.

2:10 Now the same for women. I copy the line, change the name to Female Only Demographic, and the value to Female.

> **2:25** Run it. This table has two rows. So in this sample, two households are headed by women. Filtering helps us find these answers quickly.

> **2:45** Next, I write a note: complex filter. In the next lesson, we will filter with several conditions at once. Well done!